1. Intro to Claude

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Faculty Workshop
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Today's Session

1. **Overview** →
2. **Effective Prompting** →
3. **Code and Documents** →
4. **Verifying Things** →
5. **Practice**

Overview

A Brief History

2022–2023
Online chatbots

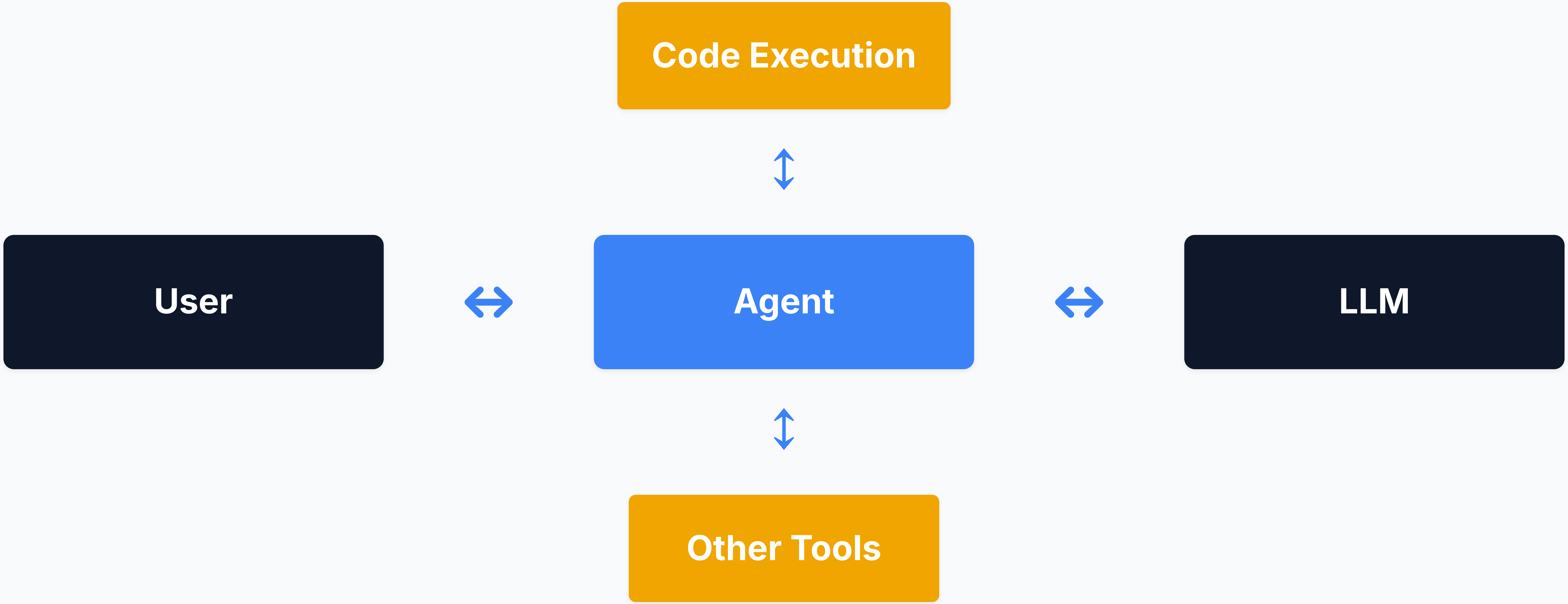
2023–2024
Code and web tools

2025
CLI tools

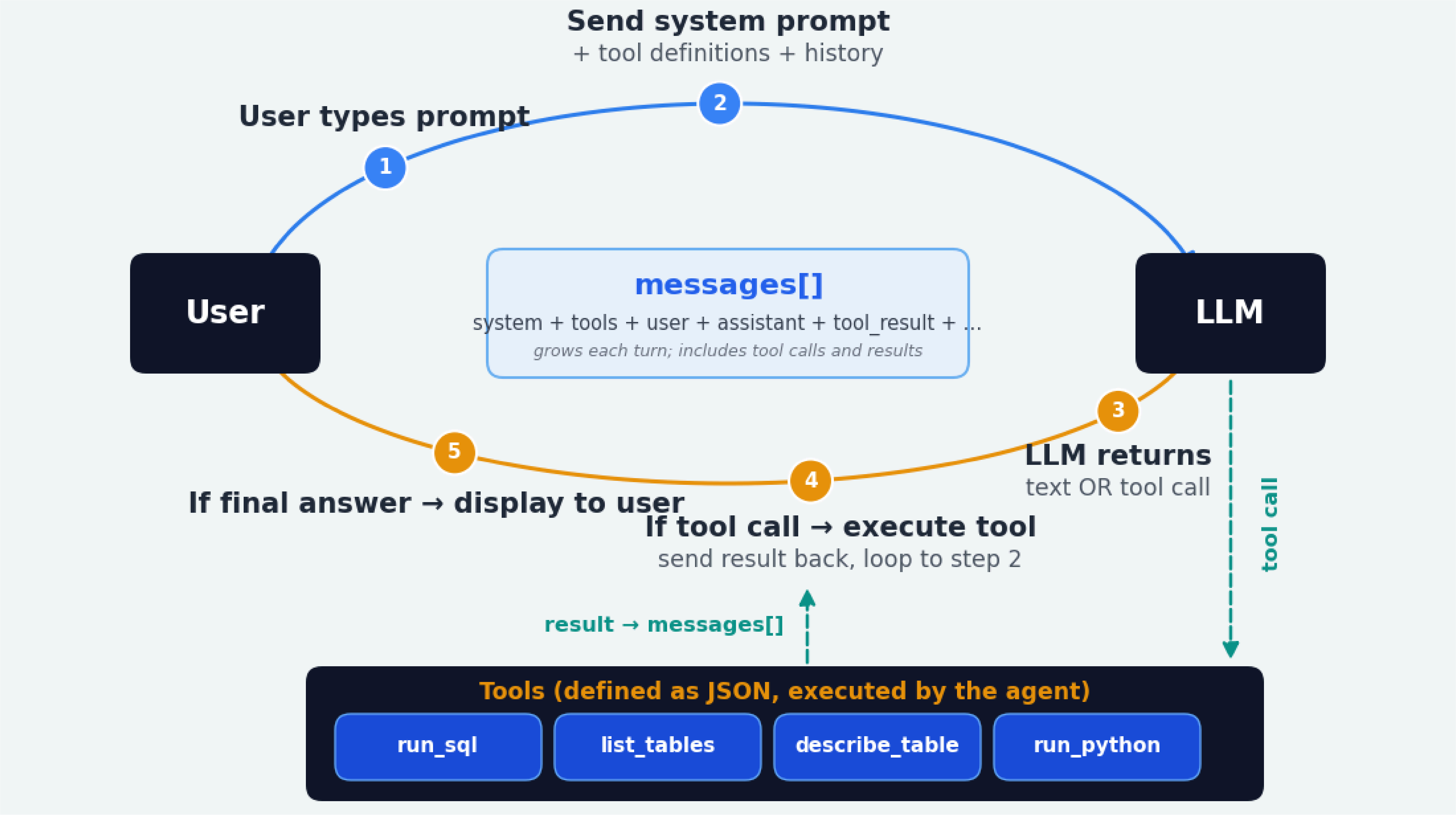
2025–2026
CLI in an app

2026
Cowork-type agents

An Agent is a Chatbot with Tools



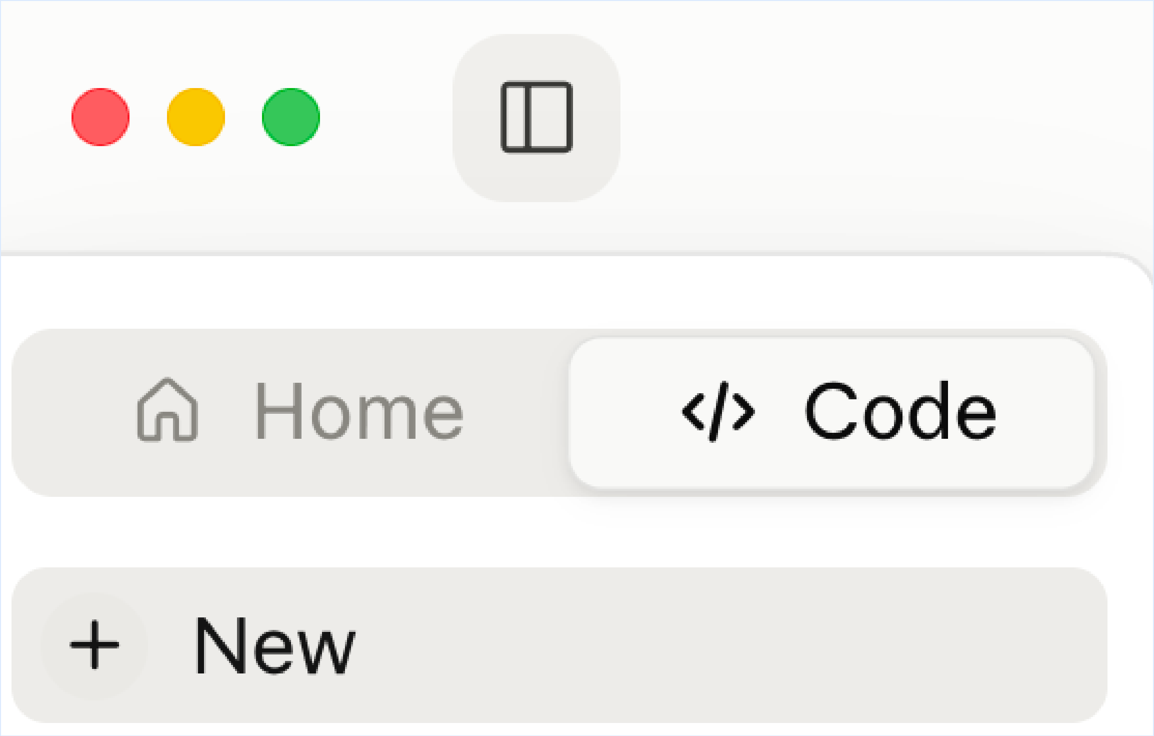
Agent in More Detail



Four Modes of Claude Desktop

Home or Code?

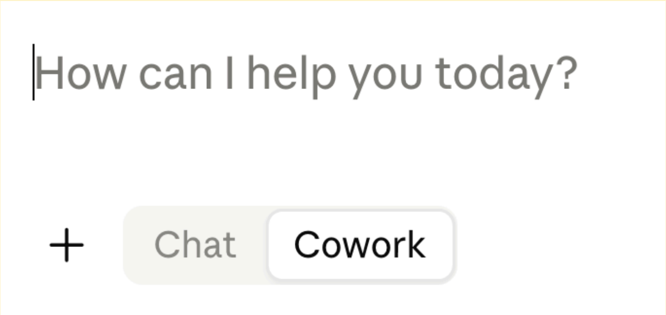
The two halves of the app, chosen at the top of the left sidebar (the square icon toggles the sidebar)



Four Modes of Claude Desktop

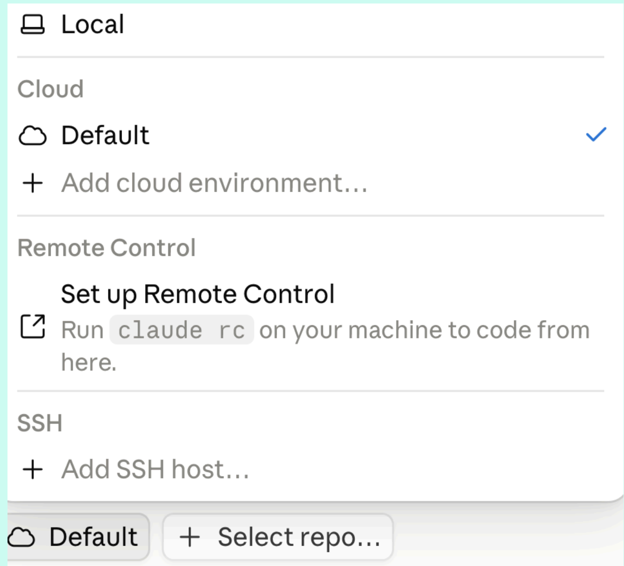
Within Home, Chat or Cowork?

Toggle in the prompt box



Within Code, Local or Cloud?

Selector at the bottom of the window



What to Use?

- You can chat in any mode
- You can execute code in any mode

Cowork > Chat on many dimensions. Requires paid account.

Code Local is best for most people, but Claude Desktop is not the best venue

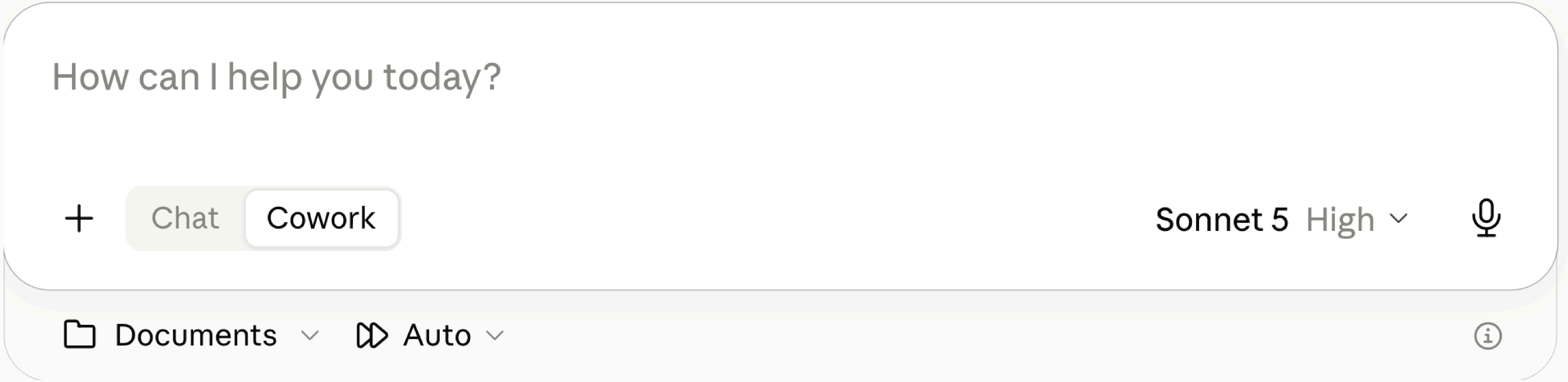
Use Cowork for now

(more later)

Code Cloud is GitHub integration, probably overkill

Configuration

- **Create New Project + Use a Folder** Give Claude access to your local files. Store project-specific instructions.
- **Auto** Automatically approve. Misleading name. Claude decides: safe → don't ask permission, possibly risky → ask permission
- **Model** I recommend Sonnet 5. Capable for anything you probably want to do. Faster and cheaper (easier on token quota).



Effective Prompting

Plan Mode

Claude Desktop has an explicit Plan mode. Or add "plan how to do this before you start."

Why It Works: Knowledge

- Planning pulls relevant knowledge into the context window
- The model answers in light of what it just retrieved, not base rates

Why It Works: Scale

- Decompose the work into subtasks
- Verify each one before the next
- Failures stay small and local

AI is a Prediction Machine, not a Rational Person

Suppose Claude has told you in the past that Method A beats Method B for some task. You ask it to do the task.

Do not assume it will use what it knows. If “A beats B” is not in the context window, and B appeared more often in its training data, it will probably use B.

This is the value of “plan before you act.” Planning surfaces “A beats B,” leading Claude to select A for the task.

Steelmanning

Don’t ask AI whether your idea is good. Ask it to make the strongest possible case that your idea is **wrong**.

Sycophancy: training on human feedback teaches that agreement gets high ratings. Ask “is my plan sound?” and Claude finds reasons it is.

The fix: “What is the strongest argument against this acquisition?”

One step further: have it play a skeptical board member, a regulator, a competitor.

Numbers Need Code

Never trust AI arithmetic. Have it write and run code.

The Problem

- “47.3%” is as plausible as “52.1%” to a next-token predictor
- Mental math, rankings, comparisons, aggregations — all unreliable
- Wrong numbers stated with complete confidence

The Fix

- Ask for Python, and have it run
- Arithmetic becomes computation, not prediction
- Code execution is built into every chatbot and Claude Code

What Not to Do

2022–2023 conventions are now useless or counterproductive.

Skip These

- “Think step by step” — reasoning models already do, in hidden tokens
- “Take a deep breath” — measurably useless now
- “CRITICAL! NEVER EVER...” — overtriggers, and gets worse results

Use With Care

- “You are an expert data analyst ...”
- Persona stacking can improve tone and structure
- Direct task framing usually wins

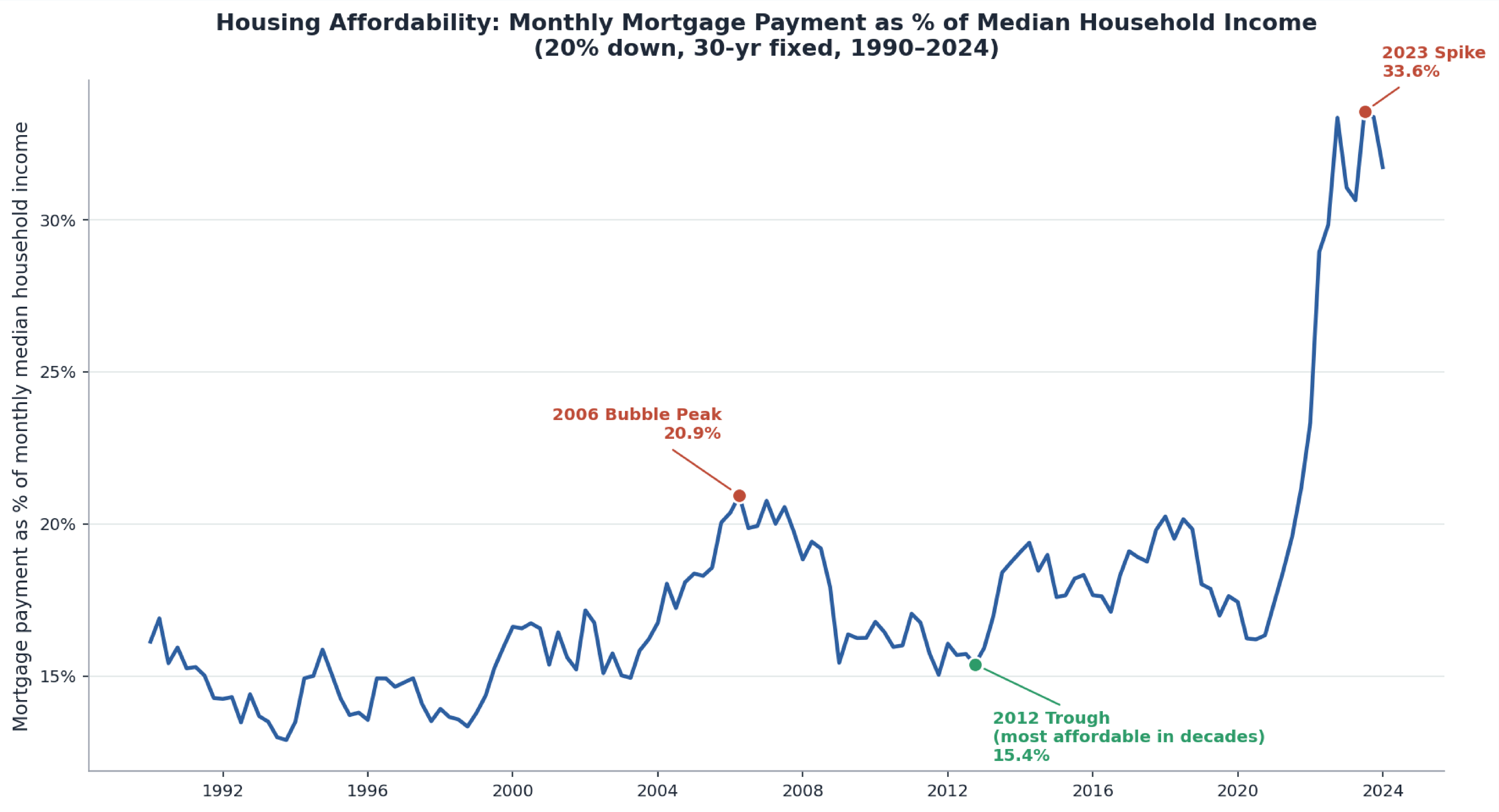
Code and Documents

Example Prompt

```
Download three FRED (Federal Reserve Economic Data) series as CSV: mortgage rate
(MORTGAGE30US), home price (MSPUS), household income (MEHOINUSA672N), 1990 through most recent.

1. MORTGAGE30US is weekly — resample to quarterly by averaging. MSPUS is already quarterly.
   MEHOINUSA672N is annual — interpolate linearly between annual observations to produce
   quarterly values.
2. Assume a 20% down payment and a 30-year loan. Compute the monthly mortgage payment on a
   median-priced home and express it as a percentage of monthly median household income.
3. Plot the affordability ratio as a line chart. Annotate three points: the 2006 bubble peak,
   the 2012 trough (most affordable in decades, as low rates met depressed prices), and the 2023
   spike.
```

Result


Housing Affordability: Monthly Mortgage Payment as % of Median Household Income
(20% down, 30-yr fixed, 1990–2024)
Mortgage payment as % of monthly median household income
30%
25%
20%
15%
1992
1996
2000
2004
2008
2012
2016
2020
2024
2006 Bubble Peak
20.9%
2012 Trough
(most affordable in decades)
15.4%
2023 Spike
33.6%

Reading, Editing, and Writing Documents

Claude natively reads **text and images**, and natively writes **text**. Everything beyond text is handled by code.

- Document reading is doc → text, then read
- Document creation is text (code) → doc
- Office document editing is ‘read and write XML (Extensible Markup Language)’

Example

Get Tesla's EPS from SEC EDGAR by quarter for the past 5 years. Plot. Create a PowerPoint deck with the plot plus discussion.

The result

[tesla.pptx]

Creating Excel Workbooks

Claude uses Python to create Excel workbooks, inserting numbers (inputs) and formulas cell-by-cell

Example Prompt

```
Build an Excel workbook containing a mortgage amortization table. Include a chart of the amount
going to interest and the amount going to principal each month.
```

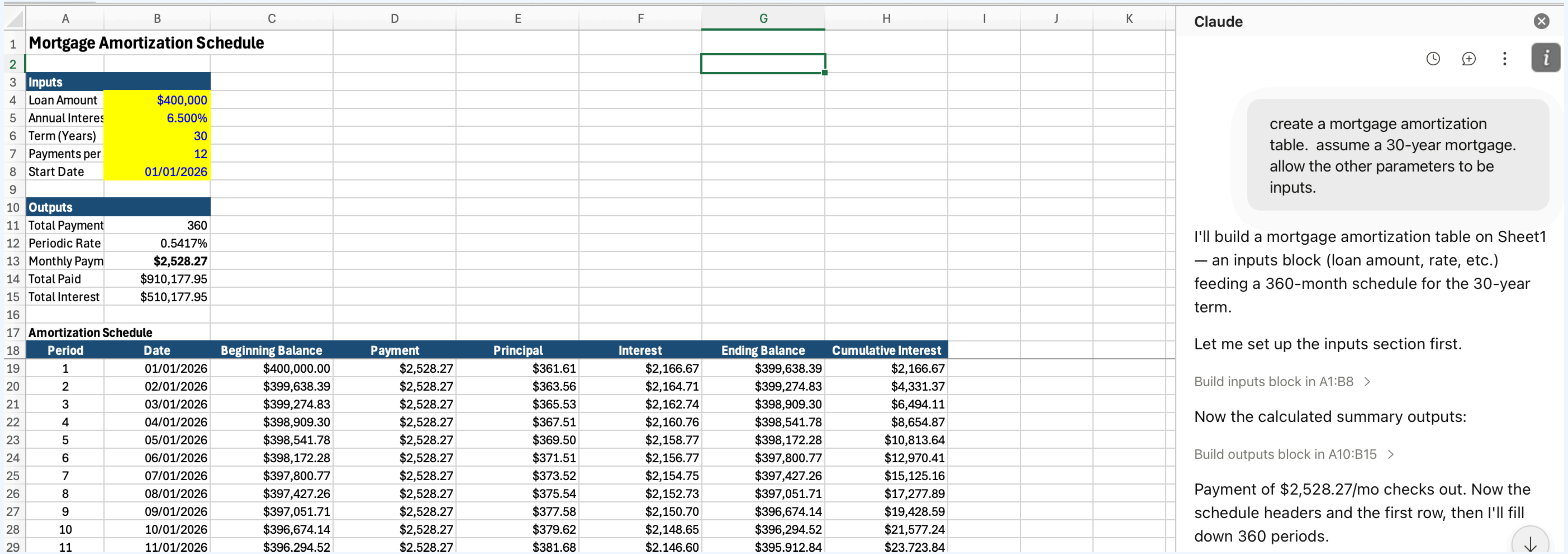
Result

[amortization.xlsx]

Claude Plugin for Excel

A chat panel inside Excel itself. Add it from `Home → Add-ins`. You will be prompted to connect to your Anthropic account (paid account).

Example



Mortgage Amortization Schedule

Inputs	
Loan Amount	$400,000
Annual Interes	6.500%
Term (Years)	30
Payments per	12
Start Date	01/01/2026

Outputs	
Total Payment	360
Periodic Rate	0.5417%
Monthly Paym	**$2,528.27**
Total Paid	$910,177.95
Total Interest	$510,177.95

Amortization Schedule

Period	Date	Beginning Balance	Payment	Principal	Interest	Ending Balance	Cumulative Interest
1	01/01/2026	$400,000.00	$2,528.27	$361.61	$2,166.67	$399,638.39	$2,166.67
2	02/01/2026	$399,638.39	$2,528.27	$363.56	$2,164.71	$399,274.83	$4,331.37
3	03/01/2026	$399,274.83	$2,528.27	$365.53	$2,162.74	$398,909.30	$6,494.11
4	04/01/2026	$398,909.30	$2,528.27	$367.51	$2,160.76	$398,541.78	$8,654.87
5	05/01/2026	$398,541.78	$2,528.27	$369.50	$2,158.77	$398,172.28	$10,813.64
6	06/01/2026	$398,172.28	$2,528.27	$371.51	$2,156.77	$397,800.77	$12,970.41
7	07/01/2026	$397,800.77	$2,528.27	$373.52	$2,154.75	$397,427.26	$15,125.16
8	08/01/2026	$397,427.26	$2,528.27	$375.54	$2,152.73	$397,051.71	$17,277.89
9	09/01/2026	$397,051.71	$2,528.27	$377.58	$2,150.70	$396,674.14	$19,428.59
10	10/01/2026	$396,674.14	$2,528.27	$379.62	$2,148.65	$396,294.52	$21,577.24
11	11/01/2026	$396,294.52	$2,528.27	$381.68	$2,146.60	$395,912.84	$23,723.84

Verifying Things

Verify Claims

The Problem

- Facts, regulations, dates, citations — stated the same way whether real or invented
- The more obscure the fact, the likelier it is fabricated

The Fix

- Have it search before answering
- Ask for sources. Then check them.

Verify Code

Consider a human assistant. Ask where they could have gone wrong.

> - They may not have understood what you wanted.
> - To perform the task, they may have had to make decisions that were different from the decisions you would have made.
> - Their Excel, code, or whatever may have had subtle bugs so it did not do what they expected.

Claude can go wrong in exactly the same ways. How do you check them?

Three Checks

Take advantage of the fact that **AI is perfectly patient**. You can probe more deeply than you would with a human assistant, and it will not be offended.

Check its understanding

Explain to me what you did and how you did it.

→

Check its decisions

Report all ways in which my instructions were ambiguous and you made decisions we have not discussed.

→

Check its code

Show me a sample case and work through the case as if by hand so I can verify your number.

Generator-Critic

Spawn a second Claude to attack the first one's output, not validate it. Subagents work well — several of them.

The Pattern

- **Generator** produces the work
- **Critic** gets: "Find every factual error, logical gap, and unsupported assumption. Do not summarize — find flaws."

Why a Separate Instance?

- A model is anchored to its own output and defends it
- A fresh instance with an adversarial mandate is not

Re-Use Tested Code

Once code runs correctly, save it. Re-use it rather than rewrite it.

The Problem with Rewriting

- Different choices each time
- Scripts that should agree may not
- Old bugs reappear

Example: Data Wrangling, Charts, Documents

Some Local Data

Extract the following from session1.zip into your project folder or just put session1.zip there and tell Claude to extract.

File	Rows	Key Columns
northwind_customers.xlsx	93	CustomerID, CompanyName, Country
northwind_orders.xlsx	500	OrderID, CustomerID, OrderDate
northwind_orderdetails.xlsx	18,976	OrderID, ProductID, UnitPrice, Quantity
northwind_products.xlsx	77	ProductID, ProductName, CategoryID
northwind_categories.xlsx	8	CategoryID, CategoryName

Merge, Filter, Aggregate, Chart, Analyze, Create

Copy this prompt into Claude:

```
I have five Northwind files in my project folder: customers, orders, order details, products,
and categories. Which product categories generated the most revenue from customers in Germany
and France? Show only categories above $10,000, sort highest to lowest, create a bar chart, and
put it in a PowerPoint deck. Add some discussion.
```

Takeaway

There is no secret to prompting. Just chat (a lot). Treat AI as an infinitely patient colleague with a poor memory. Code execution gives AI enormous power.

Homework

Watch the three-minute video academic_studio.mp4